# Forough Mahpouya (she/her)

Ph.D. Candidate in Industrial and Systems Engineering
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## RESEARCH AREAS

Causal inference and representation learning from incomplete observational data; decision-making under uncertainty; sequential experimentation and value of information; healthcare systems engineering for older adults and chronic conditions.

## EDUCATION

**Ph.D. Candidate, Industrial and Systems Engineering** 2022–present
University at Buffalo, The State University of New York, Buffalo, NY
*Expected*: June 2027
*Dissertation:* Quantitative Modeling of Health Outcomes in Patient–Caregiver Dyads in End-of-Life Care
*Advisors:* Alexander Nikolaev and Sabrina Casucci

**M.Sc., Industrial Engineering** 2017–2019
K. N. Toosi University of Technology, Tehran, Iran
*Thesis:* A Bootstrap Approach to Introduce a Robust Multivariate Control Chart
*Advisors:* Hamid Shahriari and Emad Roghanian

**B.Sc., Industrial Engineering** 2006–2010
Iran University of Science and Technology, Tehran, Iran

## PUBLICATIONS AND MANUSCRIPTS

### Refereed Journal Articles (J)

1. **Mahpouya, F.**, Burris, C. J., Paul, H., & Nikolaev, A. (2026). Maximizing the expected value of experimentation for finding top-$\kappa$ rank via aggregation of pairwise comparisons. *IISE Transactions*, 1–20.
2. Das, P. S., Casucci, S., **Mahpouya, F.**, & Nikolaev, A. (2026). Treatment effect estimation from observational data in healthcare: Study designs, inherent biases, and a propensity score-focused quality assessment scale. Accepted for publication in *Health Services and Outcomes Research Methodology*.
3. **Mahpouya, F.**, Shahriari, H., & Roghanian, E. (2022). Design of a robust $T^2$ control chart, a re-sampling approach. *Quality and Reliability Engineering International, 38*(2), 924–940.

### Manuscripts Under Review or Revision (R)

1. **Mahpouya, F.**, Das, P. S., Nikolaev, A., & Casucci, S. (2026). Use of observational causal inference in dementia management studies: A systematic literature review. Manuscript under revision for *PLOS Digital Health*.
2. Walker, R. J., Linde, S., Egede, L. E., **Mahpouya, F.**, Nikolaev, A., Fraser, R., & Martinez, J. (2026). Assessing the Healthy Eating Index and health: Psychometric properties and prediction of cardiometabolic outcomes. Manuscript under review at *The Journal of Nutrition*.
3. Rogers, A. H., Fligelman, S. J., Pester, B. D., Pielech, M., Klages, K. L., **Mahpouya, F.**, Nikolaev, A., Groenewald, C. B., Ketterl, T. G., Bevers, L. M., Dietel, K. & Palermo, T. M. (2026). Pain management and opioid use in adolescents and young adults with cancer: a qualitative analysis of patient and caregiver experiences. Manuscript under review at *Supportive Care in Cancer*.

### Works in Progress (W)

1. **Mahpouya, F.**, Nikolaev, A., & Casucci, S. (2026). Adversarial balancing for estimating the causal effect of caregiver burden on patient long-term-care transitions. Prepared for submission.

2. **Mahpouya, F.**, Casucci, S., & Sullivan, S. (2026). A population-based, bidirectional measure of caregiver risk and resilience: The Caregiver Burden–Benefit Index (CareBI). Manuscript in preparation for journal submission. Revised from the 2025 arXiv/preprint and INFORMS conference version titled "Representing caregiver burden in observational studies: Development of the Caregiver Burden Index (CareBI) using NSOC."
3. Rogers, A. H., Fligelman, S. J., Pester, B. D., Pielech, M., Klages, K. L., **Mahpouya, F.**, Nikolaev, A., Groenewald, C. B., Ketterl, T. G., Bevers, L. M., Dietel, K. E., & Palermo, T. M. (2026). Pain management and opioid use in adolescents and young adults with cancer: A quantitative analysis using Ecological Momentary Assessment (EMA). Prepared for submission.
4. Molazadeh, M., Casucci, S., **Mahpouya, F.**, Nikolaev, A., & Das, P. S. (2026). Hospice enrollment at the end of life: associations between long-term care status, functional dependence, and comorbidities. Prepared for submission.

## SELECTED RESEARCH EXPERIENCE

**Research Collaborator, CAAIR Pilot Project** 2026–present
*School of Pharmacy and Pharmaceutical Sciences, University at Buffalo, Buffalo, NY*

**Neural representation learning for polysubstance use phenotyping.** Developing and evaluating neural network methods to discover patterns of polysubstance use in linked New York State Office of Addiction Services and Supports (OASAS)–Medicaid data, comparing representations learned from categorical healthcare data with those informed by patient characteristics and substance use histories. The project is supported by the Center for Addiction and Artificial Intelligence Research (CAAIR).

**Independent Researcher, Proposal Development (I)** 2026–present
*Department of Industrial and Systems Engineering, University at Buffalo, Buffalo, NY*

1. **Opioid tapering under partial observability.** Proposal on how taper speed affects overdose and mental health outcomes when missed follow-up visits leave health problems unrecorded; feasibility review of evidence and data sources.

**Doctoral Researcher (D)** 2023–present
*Department of Industrial and Systems Engineering, University at Buffalo, Buffalo, NY*

1. **Balancing representations for continuous interventions and time-to-event outcomes.** Led theoretical development of a neural balancing framework for continuous-dose interventions with survival outcomes. Derived identification conditions, connected the learning objective to hazard modeling, and authored a technical manuscript with proofs and simulation design.
2. **Caregiver Burden–Benefit Index (CareBI).** Developed and validated the index using NHATS/NSOC data; built predictive models of adverse caregiver outcomes across caregiver profiles and service-support groups.
3. **Heterogeneous treatment effects and subgroup identification.** Authored a structured review manuscript of subgroup-identification and effect-estimation methods, with a decision map linking data structure and clinical objectives to method selection.
4. **Longitudinal analysis of diabetes and cognitive decline.** Compared cognitive decline in older adults with and without diabetes using Health and Retirement Study (HRS) data, with adversarial balancing to adjust for group differences (draft manuscript).
5. **Fairness in glomerular filtration rate prediction.** Assessed adversarial balancing for fair GFR prediction; co-authored an NIDDK data-request proposal.

**Researcher, Anomaly Detection in Inkjet Printing** 2022–2023
*Department of Industrial and Systems Engineering, University at Buffalo, Buffalo, NY*

Led theory development and preliminary design for classification-based anomaly detection from droplet pinch-off behavior; documented findings.

**Research Collaborator** 2017–2022

*Department of Industrial Engineering, K. N. Toosi University of Technology, Tehran, Iran*

Reviewed and classified multivariate population mean-vector estimators by sample count, sample size, and outlier presence; organized a literature review on degradation-data models for improving food product quality and stability.

## CONFERENCE PRESENTATIONS (C)

1. **Mahpouya, F.** (2026). Observational causal inference under clinical prevalence bias. *INFORMS 2026 Job Market Showcase*, November 1–4, San Francisco, CA.
2. **Mahpouya, F.**, Nikolaev, A., & Casucci, S. (2026). Balancing representation learning for dose-response-curve causal inference from healthcare data. *INFORMS Annual Meeting*, November 1–4, San Francisco, CA.
3. Molazadeh, M., Casucci, S., **Mahpouya, F.**, Nikolaev, A., & Das, P. S. (2026). Hospice enrollment at the end of life: associations between long-term care status, functional dependence, and comorbidities. *INFORMS Annual Meeting*, November 1–4, San Francisco, CA.
4. **Mahpouya, F.**, Nikolaev, A., & Casucci, S. (2026). Adversarial balancing for estimating the causal effect of caregiver burden on patient long-term care transitions. *IISE Annual Conference & Expo*, May 16–19, Arlington, TX.
5. **Mahpouya, F.**, Casucci, S., Sullivan, S., & Barrick, C. (2025). Representing caregiver burden in observational studies: Development of the Caregiver Burden Index (CareBI) using NSOC. *INFORMS Annual Meeting*, October 26–29, Atlanta, GA.
6. **Mahpouya, F.**, Shahriari, H., & Roghanian, E. (2020). Use of resampling techniques to design a robust Hotelling's $T^2$ control chart for limited data. *16th Iran International Industrial Engineering Conference*, Alzahra University, Tehran, Iran. Conference record
7. **Mahpouya, F.**, & Shahriari, H. (2019). A bootstrapping approach to introduce a robust multivariate control chart. *15th Iran International Industrial Engineering Conference*, Yazd University, Yazd, Iran. Conference record

## TEACHING EXPERIENCE

**Teaching Assistant and Guest Lecturer** 2022–present

*University at Buffalo, Buffalo, NY*

*Course areas:* simulation and stochastic models; production planning and control; applied probability and statistics; healthcare systems engineering; computer-integrated manufacturing; social-network behavior models.

**IE 505: Production Planning and Control** Spring 2026

**IE 477: Simulation Modeling and Analysis** Fall 2023, 2024, 2025, 2026

- Led 12 recitation sessions and two course lectures per semester in Fall 2023–2025; designed six weeks of recitation material using @RISK and Arena.

**IE 406/506: Computer-Integrated Manufacturing** Fall 2023, 2025

- Led four lab tutorials each semester using Autodesk Fusion 360 for 3D CAD modeling and manufacturing.

**IE 551: Simulation and Stochastic Models** Spring 2024, 2025

- Delivered a course lecture and held group-specific active office hours.

**IE 511: Social Network Behavior Models** Spring 2024, 2025

**IE 585: Industrial Engineering Principles and Practice in Healthcare** Fall 2024

- Delivered a guest lecture on PICO question formulation and analysis of complex observational healthcare data for interpretable evidence.

**EAS 305: Applied Probability and Statistics** Fall 2022–Summer 2023

- Led 54 recitations in Fall 2022 and 30 in Spring 2023; designed ten weeks of recitation material; delivered a Minitab lab and an exam-review lecture each semester.

## INDUSTRY AND PROFESSIONAL EXPERIENCE

**Industrial Engineer / Systems Optimizer** 2012–2016

*Arioplast, photo-frame profile manufacturing, Rasht, Iran*

- Reorganized operations and workflows, freeing two employees to staff two newly added production lines.
- Co-designed an automated weighing and pricing system that categorized products, monitored manufacturing waste in real time, and enabled immediate pricing of customer orders.
- Developed process instructions and operating manuals and trained staff on the new procedures.

**Part-Time Technical Mentor** 2010–2012; 2015–2017

*Private training institutes, Rasht and Sari, Iran*

- Designed and delivered tutorials and mentored students in ICDL, Visio, and MATLAB.

## HONORS AND AWARDS

- **University at Buffalo ISE Graduate Student Teaching Award**, 2025–2026.
- **IISE Future Faculty Fellows (3F) Program**, selected fellow, 2025–2026.
- **Exceptionally Talented Student designation**, M.Sc. program, K. N. Toosi University of Technology, 2021. Awarded after completing the M.Sc. based on graduate performance; conferred admission to the university's Ph.D. program.
- Ranked **3rd of 36** students in the M.Sc. program, K. N. Toosi University of Technology, 2019.
- Ranked in the **top 1% of 15,800 participants** in Iran's national M.Sc. entrance examination in Industrial Engineering, 2017.

## PROFESSIONAL AND ACADEMIC SERVICE

- Peer Reviewer for the *European Journal of Operational Research* and the *Journal of Nursing Measurement*.
- Contributed to the University at Buffalo's *Effective Pedagogy for Teaching Assistants* program by sharing approaches to recitation design and addressing teaching-assistant challenges through Universal Design for Learning, Spring 2025.

## METHODS AND TECHNICAL EXPERTISE

***Causal and Machine Learning Methods:*** Causal inference; balancing and adversarial representation learning; continuous-treatment effect estimation; treatment-effect heterogeneity and subgroup analysis; longitudinal data analysis; survival and hazard modeling; neural networks; fairness-aware learning.

***Statistical and Systems Methods:*** Scale development and psychometric validation; bifactor modeling; simulation and stochastic modeling; sequential experimentation and ranking; experimental design; anomaly detection; robust multivariate statistics and process control.

***Applications:*** Healthcare outcome modeling; aging and patient–caregiver systems; cardiometabolic health; manufacturing quality and process improvement.

***Programming and Research Tools:*** Python; R; MATLAB.

***Technical Software:*** Arena; @RISK; Autodesk Fusion 360; Minitab.